Lecture 2: Cooperative game theory

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Lecture 2 Outline

- Cooperative game theory
- Nash bargaining solution
- Shapley value
- Nucleolus value
- Mechanism Design
- Extra Reading:
  - Muthoo (1999). Bargaining theory with applications.

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Non-cooperative and cooperative game theory

CO What is the difference between non-cooperative and cooperative game theory?

Non-cooperative game theory and cooperative game theory are two branches of game theory that differ in how they model interactions among players.

Non-cooperative game theory assumes that players act independently, without communication or cooperation. In this model, each player makes decisions based solely on their own self-interest, without taking into account the preferences or actions of other players. Non-cooperative game theory is particularly useful for analyzing situations where there is no opportunity for negotiation or collaboration between players, and each player must make their own decisions based on incomplete information.

On the other hand, cooperative game theory assumes that players can communicate and collaborate in order to achieve mutual gains. In this model, players work together to create agreements or alliances that benefit everyone involved. Cooperative game theory is particularly useful for analyzing situations where players have the opportunity to make binding agreements, or where they can coordinate their actions to achieve a common goal.

In summary, non-cooperative game theory assumes that players act independently and competitively, while cooperative game theory assumes that players can communicate and collaborate to achieve mutual benefits.



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For example

- In a prisoner's dilemma game:

		Player 2	
		Left	Right
Player 1	Up	(2,6)	(4,5)
	Down	(1,8)	(3,7)

- In a coordination game:

		Player 2	
		Left	Right
Player 1	Up	(8,7)	(4,3)
	Down	(2,1)	(6,5)

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Main focus of cooperative game

- A "reasonable", and, hopefully, implementable solution to distribute benefits or cost for everyone.
- Elevator problem: how much should each floor pay?



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Properties

- Weak Pareto
- Monotonicity
- Symmetry
- Scale Invariance
- Nash Independence
- ...



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Properties

In cooperative game theory, the following are the properties that are often used to evaluate and compare different solutions to a cooperative game:

1. Weak Pareto: A solution satisfies the weak Pareto property if there is no other feasible solution that is worse for all players and strictly better for at least one player. In other words, a solution is weak Pareto optimal if no player can be made worse off without making someone else better off.
2. Symmetry: A solution satisfies the symmetry property if players who are "equivalent" in some sense receive the same allocation. For example, if two players have the same input to the game and contribute equally to the output, then they should receive the same allocation.
3. Scale invariance: A solution satisfies the scale invariance property if it is invariant under linear transformations of the game, such as multiplying all payoffs by a constant factor. This means that the solution should not be affected by changes in the units of measurement or the overall magnitude of the payoffs.
4. Nash Independence: A solution satisfies the Nash independence property if it does not depend on the order in which the players make their demands. In other words, a solution is Nash independent if the outcome is the same regardless of the order in which the players negotiate.
5. Monotonicity: A solution satisfies the monotonicity property if a player can never be worse off by increasing their contribution to the game. This means that the allocation should always be at least as good as the allocation that would result if the player contributed less.

Note that different solutions to a cooperative game may satisfy different combinations of these properties, and the choice of which properties to impose depends on the specific context and goals of the game.

Regenerate response

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Nash bargaining solution

- Assume u_i to be player i's utility if negotiation succeeds, d_i to be player i's utility if negotiation fails.
- The Nash bargaining solution (NBS) of the bargaining situation is the unique pair of utilities that solves the following maximization problem

$$max\ (u_1 - d_1)(u_2 - d_2)$$

- Basically, instead of answering "what would be a (most) likely negotiation outcome", we are now discussing "what should be the socially optimal negotiation outcome".

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Risk-neutral Nash bargaining solution

- (Split-the-difference-rule) Suppose that both players have the most simple linear utility functions $u_s = s$, then each player earns a share of $\frac{1+d_1-d_2}{2}$ and $\frac{1-d_1+d_2}{2}$, respectively.
- Proof: suppose that player 2 gets a share of x

$$max\ (1-x-d_1)(x-d_2)$$

$$FOD : (1-x-d_1)-(x-d_2) = 1-2x-d_1+d_2 = 0$$

$$\Longrightarrow 1-x = \frac{1+d_1-d_2}{2}\ and\ x = \frac{1-d_1+d_2}{2}$$

- How to interpret the NBS here?

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Risk-averse Nash bargaining solution

- Suppose that both players have concave utility functions $u_1(s) = s^\alpha$ and $u_2(s) = s^\beta$ $(0 < \alpha < 1$ and $0 < \beta < 1)$ and zero disagreement points. Then each player has a share of $\frac{\alpha}{\alpha+\beta}$ and $\frac{\beta}{\alpha+\beta}$, respectively.
- Proof: suppose that player 2 gets a share of x

$$max\ (1-x)^\alpha x^\beta$$

$$FOD : (1-x)^\alpha \beta x^{\beta-1} - \alpha(1-x)^{\alpha-1}x^\beta = 0$$

$$\implies (1-x)^\alpha \beta x^{\beta-1} = \alpha(1-x)^{\alpha-1}x^\beta$$

$$\implies (1-x)\beta = \alpha x$$

$$\implies 1 - x = \frac{\alpha}{\alpha+\beta}\ and\ x = \frac{\beta}{\alpha+\beta}$$

- How to interpret the NBS here?

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Asymmetric Nash bargaining solution

- The asymmetric Nash bargaining solution of the bargaining situation is the unique pair of utilities that solves the following maximization problem

$$max\ (u_1 - d_1)^{\tau}(u_2 - d_2)^{1-\tau}$$

- (Split-the-difference-rule) Suppose that both players have the simplest linear utility functions $u_s = s$, then each player earns a share of $(1-\tau)d_1 + \tau(1-d_2)$ and $(1-\tau)(1-d_1) + \tau d_2$, respectively.
- How to interpret the NBS here?



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Asymmetric Nash bargaining solution

- Proof: suppose that player 2 gets a share of x

$$max\ (1-x-d_1)^{\tau}(x-d_2)^{1-\tau}$$

$$FOD:\ (1-x-d_1)^{\tau}(1-\tau)(x-d_2)^{-\tau}-\tau(1-x-d_1)^{\tau-1}(x-d_2)^{1-\tau}=0$$

$$\Longrightarrow (1-\tau)(\frac{1-x-d_1}{x-d_2})^{\tau}=\tau(\frac{1-x-d_1}{x-d_2})^{\tau-1}$$

$$\Longrightarrow \frac{1-x-d_1}{x-d_2}=\frac{\tau}{1-\tau}$$

$$\Longrightarrow 1-\tau-(1-\tau)x-(1-\tau)d_1=\tau x-\tau d_2$$

$$\Longrightarrow x=(1-\tau)(1-d_1)+\tau d_2$$

$$\Longrightarrow 1-x=(1-\tau)d_1+\tau(1-d_2)$$



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Three players: one principal and two agents

- Now, you (P) are working on a project that requires at least two people. You can hire at least one of your two friends (A_1 and A_2) to work on the project.



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- You work with A_1, your project yields a profit of $\frac{1}{2}$.



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$V_{P2} = \frac{2}{3}$

- You work with A_2, your project yields a profit of $\frac{2}{3}$.



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$V_{P12} = 1$

- You work with both, your project yields a profit of 1.



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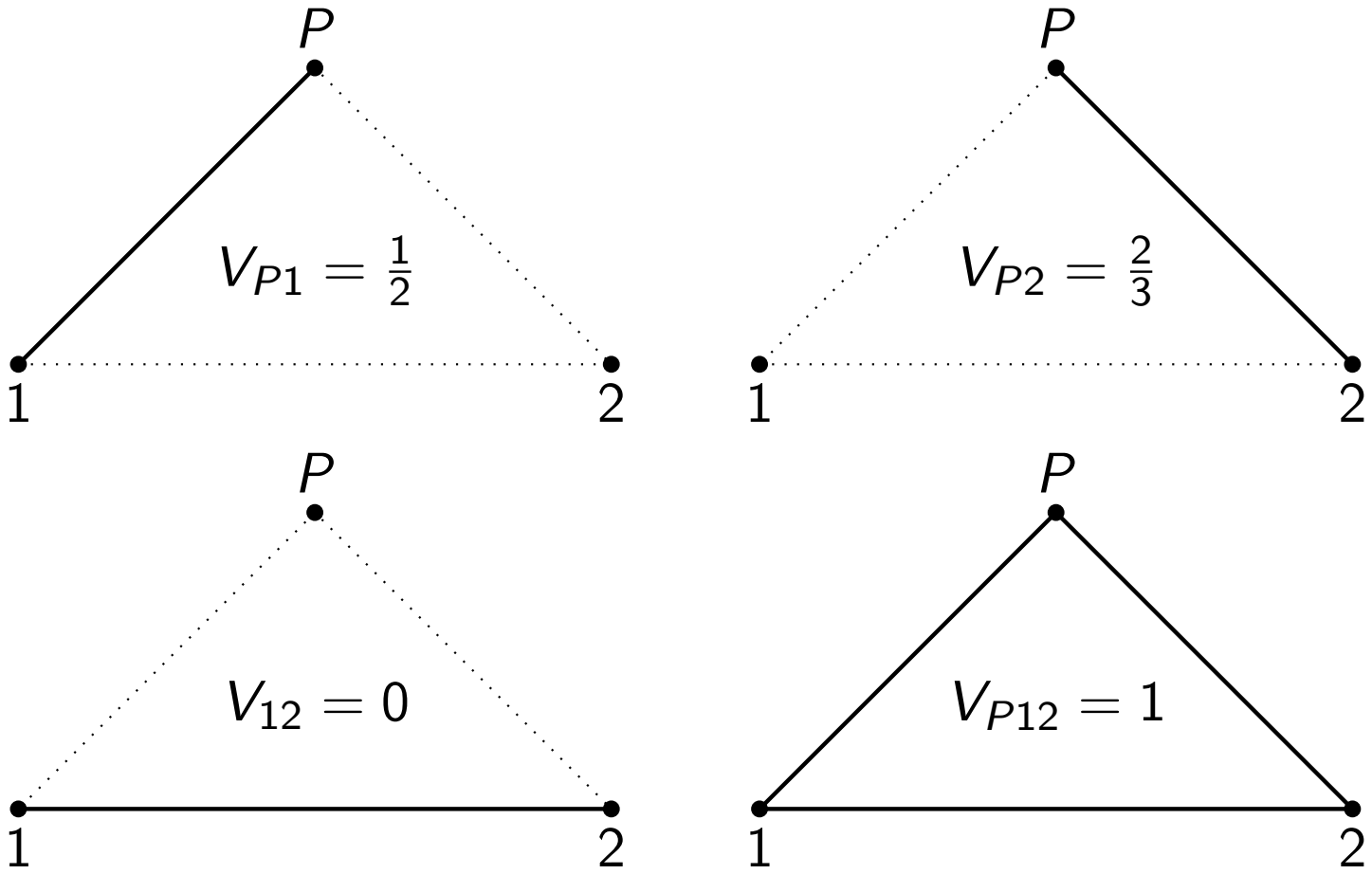
$V_{12} = 0$

- Without you, there is no business.



- So: how much "SHOULD" each of you get?

In summary



- Take a look at this issue. How do we find "reasonable" distributions between these three players?
- There can be more than one way to do so. We introduce two values for this purpose.

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Shapley value

- The table shows the **marginal contributions** players make when they enter the project in different orders.

$\Longrightarrow$	P	1	2
$P12$	0	$\frac{1}{2}$	$\frac{1}{2}$
$P21$	0	$\frac{1}{3}$	$\frac{2}{3}$
$1P2$	$\frac{1}{2}$	0	$\frac{1}{2}$
$12P$	1	0	0
$2P1$	$\frac{2}{3}$	$\frac{1}{3}$	0
$21P$	1	0	0
average	$\frac{19}{36}$	$\frac{7}{36}$	$\frac{10}{36}$

- The average of the "added value" is the Shapley Value: $(\frac{19}{36}, \frac{7}{36}, \frac{10}{36})$.

Notice: for consistency, always express the outcome in the order of (P, A_1, A_2).

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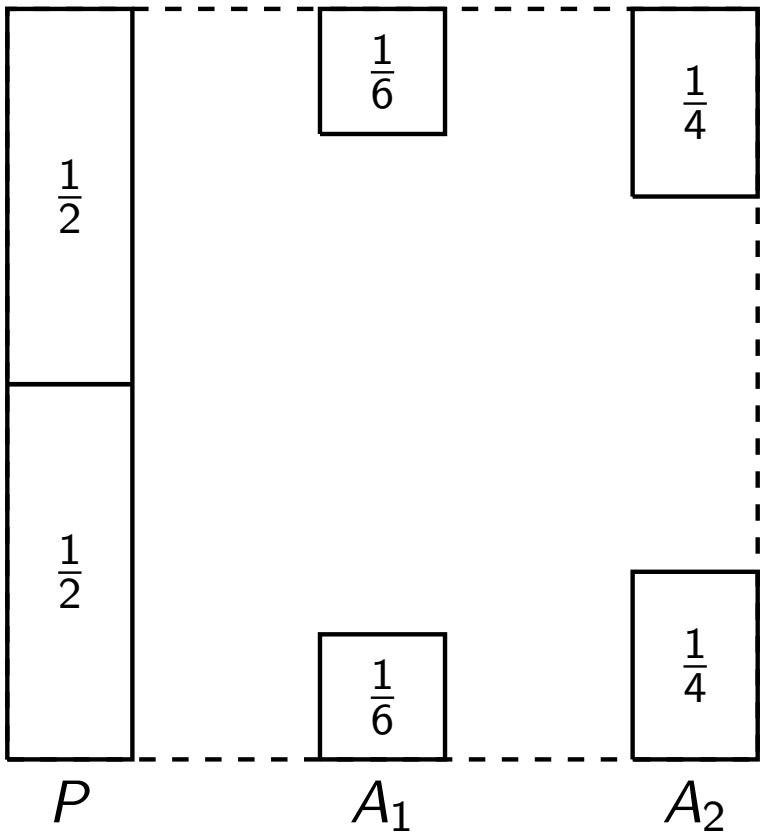
Nucleolus Value

- Or, let us think about how much each individual can take away from the firm.
- Without agent 1, the maximum profit is $V_{P2} = \frac{2}{3}$, the firm loses $1 - V_{P2} = \frac{1}{3}$.
- Without agent 2, the maximum profit is $V_{P1} = \frac{1}{2}$, the firm loses $1 - V_{P1} = \frac{1}{2}$.
- Without the principal, the maximum profit is $V_{12} = 0$, the firm loses $1 - V_{12} = 1$.

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Nucleolus Value

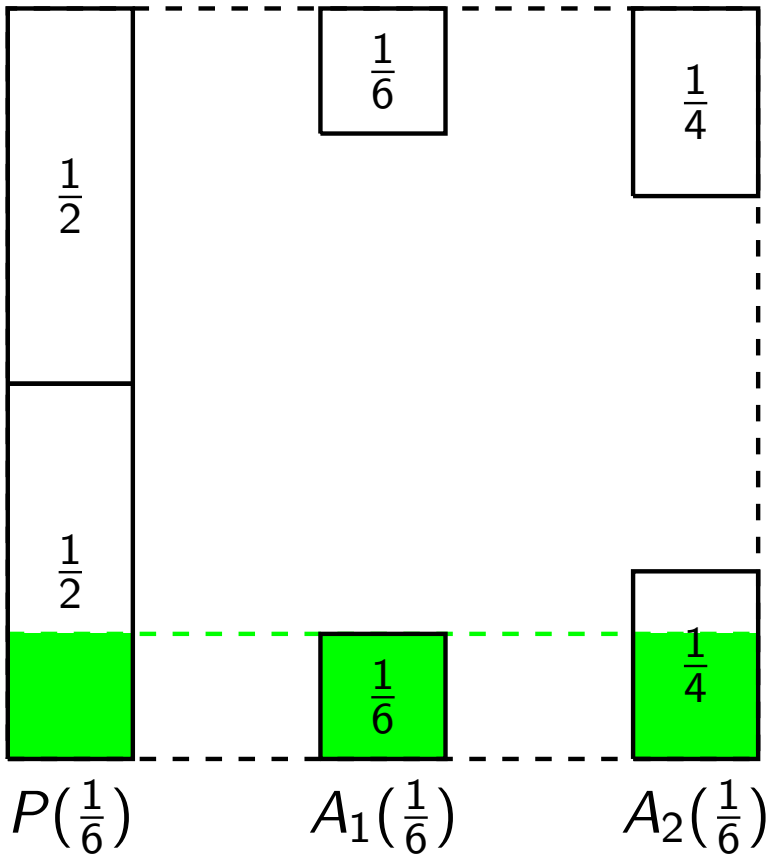
- Imagine the length of the pipes in the graph represents the "threat". Cut each pipe in half. We are now distributing the total profit to the three players, according to how "powerful" they are.



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Nucleolus Value

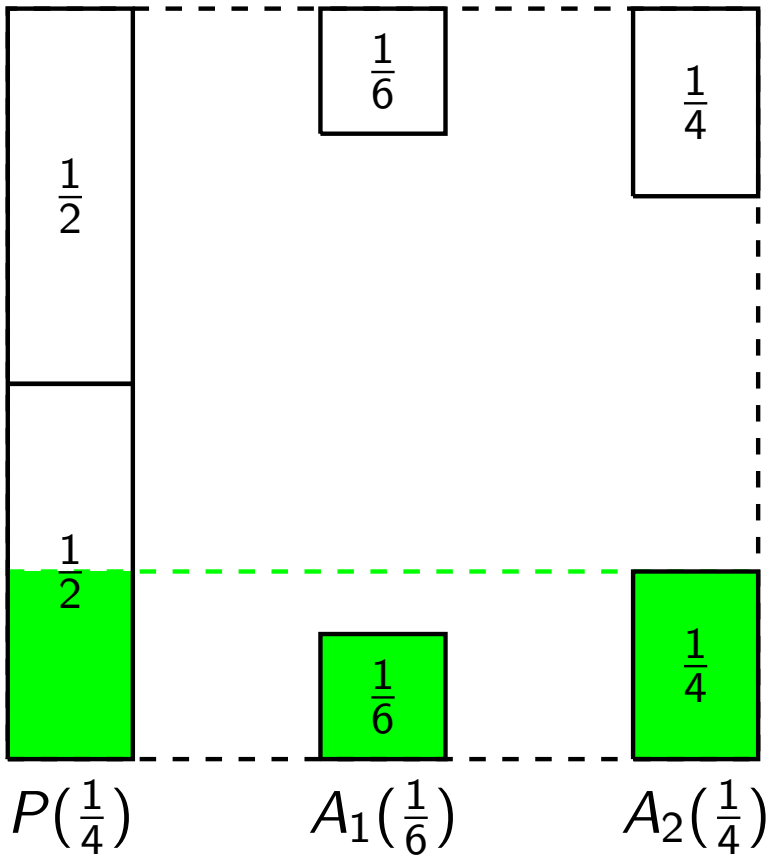
- Now, imagine the total profit 1 is like water to be poured into the container. First the lowest pipe is filled.



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Nucleolus Value

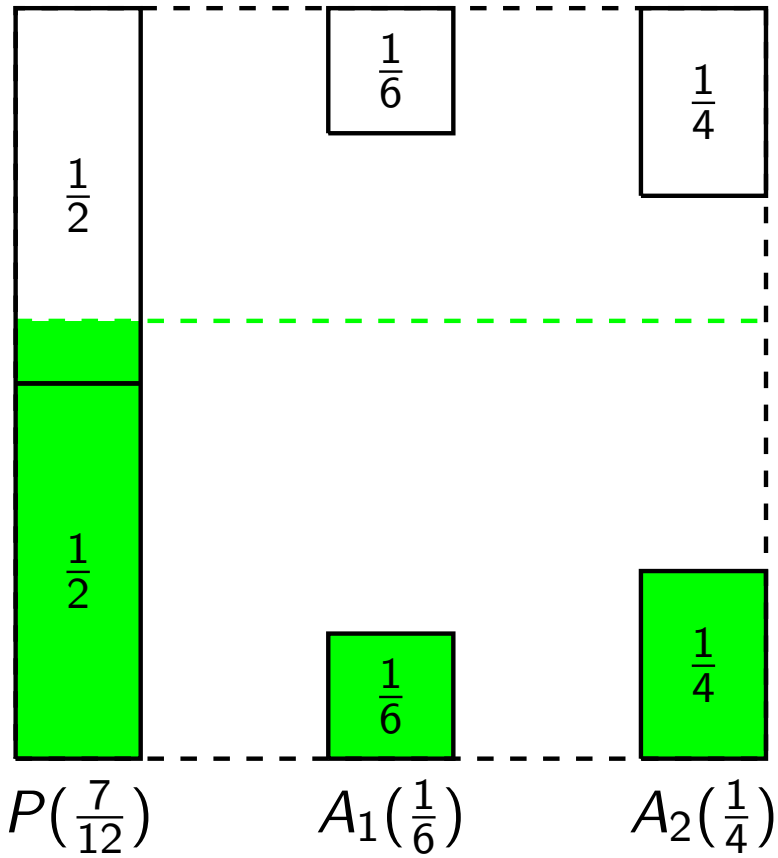
- Then the second lowest pipe is filled.



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Nucleolus Value

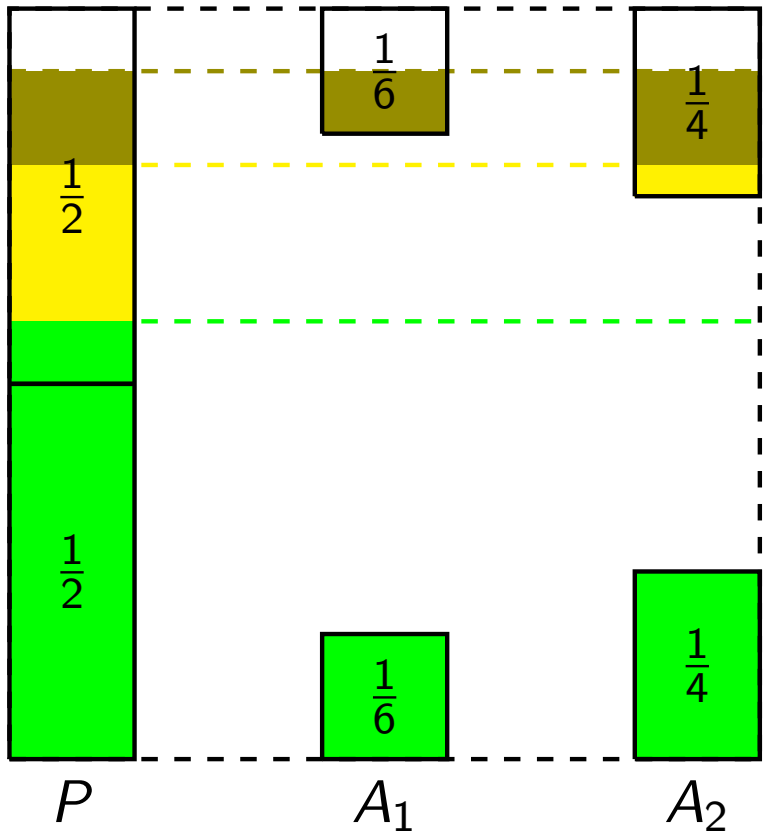
- Keep pouring the water until there is no more remaining.
- Now we have the result of $(\frac{7}{12}, \frac{1}{6}, \frac{1}{4})$, which is the Nucleolus Value.



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Nucleolus Value

- Suppose that we still have more to distribute...



Nucleolus Value

- So, why cut the "pipes" in half? What about cutting them in, say, one-third and two-thirds?
- Long story in short, you can. Notice that we have different numerical results in Shapley and Nucleolus values; having other "reasonable" distributions is normal. If we cut the pipes in different proportions, we will have other "Nucleolus-alike" values. The intuitions will be similar.



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Mechanism Design

- From cooperative game theory to non-cooperative game theory, we need something called **Mechanism Design**, to implement the "reasonable" distribution.
- In mechanism design, we induce instead of enforce.
- A simple example: suppose that we wish two players to divide a pie as equal (half-to-half) as possible.
- What is enforce?
  - cut the pie for them
  - grab their hands to cut the pie
  - threaten them to divide equally or we will kill them
  - ...
- What is induce?
  - Make the rule that one player will cut the pie and the other player will pick.



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Mechanism Design

- The main point is to create a rule that makes people do what we want them do do on their own.
- Let's look back at the Elevator problem we talked about earlier. We can't order each building around. Instead, we can suggest some properties in distributing for them to consider and set up simple but effective rules for their negotiations.



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Thank you



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